

RAJALAKSHMI ENGINEERING COLLEGE

COMPUTER NETWORKS

EXPERIMENT 8

NMAP LIVE HOST DISCOVERY

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Nmap Live Host Discovery

Learn how to use Nmap to discover live hosts using ARP scan, ICMP scan, and TCP/UDP ping scan.

📶 🟢 ⌚ 120 min 👤 233,848 🗨️ 1

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Room completed (100%)

Task 1 Introduction

Task 2 Subnetworks

Task 3 Enumerating Targets

Task 4 Discovering Live Hosts

Task 5 Nmap Host Discovery Using ARP

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Task 7 Nmap Host Discovery Using TCP and UDP

Task 8 Using Reverse-DNS Lookup

Task 9 Summary

Task 1 Introduction

When we want to target a network, we want to find an efficient tool to help us handle repetitive tasks and answer the following questions:

1. Which systems are up?
2. What services are running on these systems?

The tool that we will rely on is Nmap. The first question about finding live computers is answered in this room. This room is the first in a series of four rooms dedicated to Nmap. The second question about discovering running services is answered in the next Nmap rooms that focus on port-scanning.

This room is the first of four in this Nmap series. These four rooms are also part of the Network Security module.

1. [Nmap Live Host Discovery](#)
2. [Nmap Basic Port Scans](#)
3. [Nmap Advanced Port Scans](#)
4. [Nmap Post Port Scans](#)

This room explains the steps that Nmap carries out to discover the systems that are online before port-scanning. This stage is crucial because trying to port-scan offline systems will only waste time and create unnecessary noise on the network.

We present the different approaches that Nmap uses to discover live hosts. In particular, we cover:

1. ARP scan: This scan uses ARP requests to discover live hosts
2. ICMP scan: This scan uses ICMP requests to identify live hosts
3. TCP/UDP ping scan: This scan sends packets to TCP ports and UDP ports to determine live hosts.

We also introduce two scanners, `arp-scan` and `masscan`, and explain how they overlap with part of Nmap's host discovery.

As already mentioned, starting with this room, we will use Nmap to discover systems and services actively. Nmap was created by Gordon Lyon (Fyodor), a network security expert and open source programmer. It was released in 1997. Nmap, short for Network Mapper, is free, open-source software released under GPL license. Nmap is an industry-standard tool for mapping networks, identifying live hosts, and discovering running services. Nmap's scripting engine can further extend its functionality, from fingerprinting services to exploiting

- 1 Enumerate targets
- 2 Discover live hosts
- 3 Reverse-DNS lookup
- 4 Scan ports
- 5 Detect versions
- 6 Detect OS
- 7 Traceroute
- 8 Scripts
- 9 Write output

Answer the questions below

Some of these questions will require the use of a static site to answer the task questions, while others require the use of the AttackBox.

No answer provided

Correct Answer

Task 2 Subnetworks

Let's review a couple of terms before we move on to the main tasks. A *network segment* is a group of computers connected using a shared medium. For instance, the medium can be the Ethernet switch or WiFi access point. In an IP network, a *subnetwork* is usually the equivalent of one or more network segments connected together and configured to use the same router. The *network segment* refers to a physical connection, while a *subnetwork* refers to a logical connection.

[View Site](#)

In the following network diagram, we have four network segments or subnetworks. Generally speaking, your system would be connected to one of these network segments/subnetworks. A subnetwork, or simply a subnet, has its own IP address range and is connected to a more extensive network via a router. There might be a firewall enforcing security policies depending on each network.



The figure above shows two types of subnets:

- Subnets with `/16`, which means that the subnet mask can be written as `255.255.0.0`. This subnet can have around 65 thousand hosts.
- Subnets with `/24`, which indicates that the subnet mask can be expressed as `255.255.255.0`. This subnet can have around 250 hosts.

You might want to refer to Task 2 in the [Intro to LAN](#) room if you need to learn more about subnetting.

As part of active reconnaissance, we want to discover more information about a group of hosts or about a subnet. If you are connected to the same subnet, you would expect your scanner to rely on `ARP` (Address Resolution Protocol) queries to discover live hosts. An `ARP` query aims to get the hardware address (MAC address) so that communication over the link layer becomes possible; however, we can use this to infer that the host is online. (We revisit link layer in Task 4.)

If you are in Network A, you can use `ARP` only to discover the devices within that subnet (10.1.100.0/24). Suppose you are connected to a subnet different from the subnet of the target system(s). In that case, all packets generated by your scanner will be routed via the default gateway (router) to reach the systems on another subnet; however, the `ARP` queries won't be routed and hence cannot cross the subnet router. `ARP` is a link layer protocol, and `ARP` packets are bound to their subnet.

Click on the "View Site" button to start the network simulator. We will use this simulator to answer the questions in tasks 2, 4, and 5.

Answer the questions below

Send a packet with the following:

Send Packet

From:

computer1

To:

computer1

Packet Type:

arp_request

Data:

computer6

Send Packet

- From computer1
- To computer1 (to indicate it is broadcast)
- Packet Type: "ARP Request"
- Data: computer6 (because we are asking for computer6 MAC address using ARP Request)

How many devices can see the ARP Request?

4

✓ Correct Answer

🔍 Hint

Did computer6 receive the ARP Request? (Y/N)

N

✓ Correct Answer

Send a packet with the following:

Send Packet

From:

computer4

To:

computer4

Packet Type:

arp_request

Data:

computer6

Send Packet

- From computer4
- To computer4 (to indicate it is broadcast)
- Packet Type: "ARP Request"
- Data: computer6 (because we are asking for computer6 MAC address using ARP Request)

How many devices can see the ARP Request?

4

✓ Correct Answer

🔍 Hint

Did computer6 reply to the ARP Request? (Y/N)

Y

✓ Correct Answer

Task 3 Enumerating Targets

We mentioned the different *techniques* we can use for scanning in Task 1. Before we explain each in detail and put it into use against a live target, we need to specify the targets we want to scan. Generally speaking, you can provide a list, a range, or a subnet. Examples of target specification are:

- list: `MACHINE_IP scanme.nmap.org example.com` will scan 3 IP addresses.
- range: `10.11.12.15-20` will scan 6 IP addresses: `10.11.12.15`, `10.11.12.16`, ..., and `10.11.12.20`.
- subnet: `MACHINE_IP/30` will scan 4 IP addresses.

You can also provide a file as input for your list of targets, `nmap -iL list_of_hosts.txt`.

If you want to check the list of hosts that `Nmap` will scan, you can use `nmap -sL TARGETS`. This option will give you a detailed list of the hosts that `Nmap` will scan without scanning them; however, `Nmap` will attempt a reverse-DNS resolution on all the targets to obtain their names. Names might reveal various information to the pentester. (If you don't want `Nmap` to the DNS server, you can add `-n`.)

Launch the AttackBox using the Start AttackBox button, open the terminal when the AttackBox is ready, and use `Nmap` to answer the following.

Answer the questions below

What is the first IP address Nmap would scan if you provided `10.10.12.13/29` as your target?

✓ Correct Answer

🔍 Hint

How many IP addresses will Nmap scan if you provide the following range `10.10.0-255.101-125`?

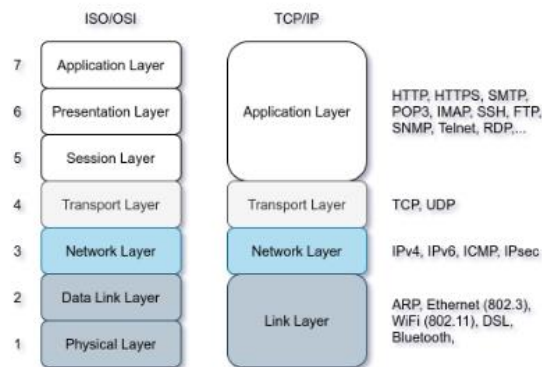
✓ Correct Answer

🔍 Hint

Task 4 Discovering Live Hosts

Let's revisit the TCP/IP layers shown in the figure next. We will leverage the protocols to discover the live hosts. Starting from bottom to top, we can use:

- ARP from Link Layer
- ICMP from Network Layer
- TCP from Transport Layer
- UDP from Transport Layer



Before we discuss how scanners can use each in detail, we will briefly review these four protocols. ARP has one purpose: sending a frame to the broadcast address on the network segment and asking the computer with a specific IP address to respond by providing its MAC (hardware) address.

ICMP has many types. ICMP ping uses Type 8 (Echo) and Type 0 (Echo Reply).

If you want to ping a system on the same subnet, an ARP query should precede the ICMP Echo.

Although TCP and UDP are transport layers, for network scanning purposes, a scanner can send a specially-crafted packet to common TCP or UDP ports to check whether the target will respond. This method is efficient, especially when ICMP Echo is blocked.

If you have closed the network simulator, click on the "View Site" button in Task 2 to display it again.

Send a packet with the following:

- From computer1
- To computer3
- Packet Type: "Ping Request"

What is the type of packet that computer1 sent before the ping?

ARP Request

✓ Correct Answer

What is the type of packet that computer1 received before being able to send the ping?

ARP Response

✓ Correct Answer

How many computers responded to the ping request?

1

✓ Correct Answer

Send a packet with the following:

- From computer2
- To computer5
- Packet Type: "Ping Request"

What is the name of the first device that responded to the first ARP Request?

router

✓ Correct Answer

What is the name of the first device that responded to the second ARP Request?

computer5

✓ Correct Answer

Send another Ping Request. Did it require new ARP Requests? (Y/N)

N

✓ Correct Answer

Task 5 Nmap Host Discovery Using ARP

How would you know which hosts are up and running? It is essential to avoid wasting our time port-scanning an offline host or an IP address not in use. There are various ways to discover online hosts. When no host discovery options are provided, Nmap follows the following approaches to discover live hosts:

1. When a *privileged* user tries to scan targets on a local network (Ethernet), Nmap uses ARP requests. A privileged user is `root` or a user who belongs to `sudoers` and can run `sudo`.
2. When a *privileged* user tries to scan targets outside the local network, Nmap uses ICMP echo requests, TCP ACK (Acknowledge) to port 80, TCP SYN (Synchronize) to port 443, and ICMP timestamp request.
3. When an *unprivileged* user tries to scan targets outside the local network, Nmap resorts to a TCP 3-way handshake by sending SYN packets to ports 80 and 443.

Nmap, by default, uses a ping scan to find live hosts, then proceeds to scan live hosts only. If you want to use Nmap to discover online hosts without port-scanning the live systems, you can issue `nmap -sn TARGETS`. Let's dig deeper to gain a solid understanding of the different techniques used.

ARP scan is possible only if you are on the same subnet as the target systems. On an Ethernet (802.3) and WiFi (802.11), you need to know the MAC address of any system before you can communicate with it. The MAC address is necessary for the link-layer header; the header contains the source MAC address and the destination MAC address among other fields. To get the MAC address, the OS sends an ARP query. A host that replies to ARP queries is up. The ARP query only works if the target is on the same subnet as yourself, i.e., on the same Ethernet/WiFi. You should expect to see many ARP queries generated during a Nmap scan of a local network. If you want Nmap only to perform an ARP scan without port-scanning, you can use `nmap -PR -sn TARGETS`, where `-PR` indicates that you only want an ARP scan. The following example shows Nmap using ARP for host discovery without any port scanning. We run `nmap -PR -sn MACHINE_IP/24` to discover all the live systems on the same subnet as our target machine.

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PR -sn 10.10.210.6/24

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-02 07:12 BST
Nmap scan report for ip-10-10-210-75.eu-west-1.compute.internal (10.10.210.75)
Host is up (0.00013s latency).
MAC Address: 02:83:75:3A:F2:89 (Unknown)
Nmap scan report for ip-10-10-210-100.eu-west-1.compute.internal (10.10.210.100)
Host is up (-0.100s latency).
MAC Address: 02:63:D0:1B:2D:CD (Unknown)
Nmap scan report for ip-10-10-210-165.eu-west-1.compute.internal (10.10.210.165)
Host is up (0.00025s latency).
MAC Address: 02:59:79:4F:17:87 (Unknown)
Nmap scan report for ip-10-10-210-6.eu-west-1.compute.internal (10.10.210.6)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.12 seconds
```

02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.0? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.1? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.2? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.3? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.4? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.5? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	ARP Announcement for 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.7? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.8? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.9? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.10? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.11? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.12? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.13? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.14? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.15? Tell 10.10.210.6
02:ba:eb:d6:18:2b	Broadcast	ARP	who has 10.10.210.16? Tell 10.10.210.6

Address Resolution Protocol: Protocol Packets: 1207 · Displayed: 512 (42.4%) Profile: Default

If you have closed the network simulator, click on the "Visit Site" button in Task 2 to display it again.

Answer the questions below

We will be sending broadcast ARP Requests packets with the following options:

- From computer1
- To computer1 (to indicate it is broadcast)
- Packet Type: "ARP Request"
- Data: try all the possible eight devices (other than computer1) in the network: computer2, computer3, computer4, computer5, computer6, switch1, switch2, and router.

How many devices are you able to discover using ARP requests?

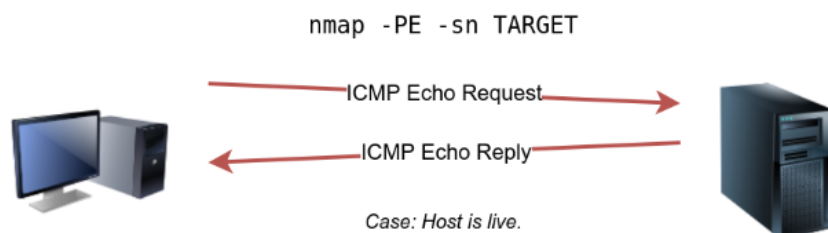
3

✓ Correct Answer

Task 6 Nmap Host Discovery Using ICMP

We can ping every IP address on a target network and see who would respond to our **ping** (ICMP Type 8/Echo) requests with a ping reply (ICMP Type 0). Simple, isn't it? Although this would be the most straightforward approach, it is not always reliable. Many firewalls block ICMP echo; new versions of MS Windows are configured with a host **firewall** that blocks ICMP echo requests by default. Remember that an **ARP** query will precede the ICMP request if your target is on the same subnet.

To use ICMP echo request to discover live hosts, add the option **-PE** (Remember to add **-sn** if you don't want to follow that with a port scan.) As shown in the following figure, an ICMP echo scan works by sending an ICMP echo request and expects the target to reply with an ICMP echo reply if it is online.

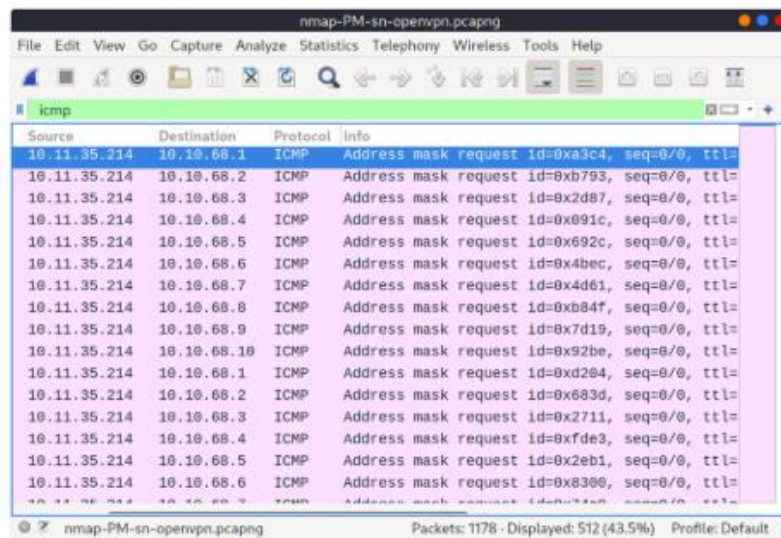


In the example below, we scanned the target's subnet using **nmap -PE -sn MACHINE_IP/24**. This scan will send ICMP echo packets to every IP address on the subnet. Again, we expect live hosts to reply; however, it is wise to remember that many firewalls block ICMP. The output below shows the result of scanning the virtual machine's class C subnet using **sudo nmap -PE -sn MACHINE_IP/24** from the AttackBox.

```
Pentester Terminal
pentester@TryHackMe$ sudo nmap -PE -sn 10.10.68.220/24

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-02 10:16 BST
Nmap scan report for ip-10-10-68-50.eu-west-1.compute.internal (10.10.68.50)
Host is up (0.00017s latency).
MAC Address: 02:95:36:71:58:87 (Unknown)
Nmap scan report for ip-10-10-68-52.eu-west-1.compute.internal (10.10.68.52)
Host is up (0.00017s latency).
MAC Address: 02:48:E8:BF:78:E7 (Unknown)
Nmap scan report for ip-10-10-68-77.eu-west-1.compute.internal (10.10.68.77)
Host is up (-0.100s latency).
```


Although we didn't get any reply and could not figure out which hosts are online, it is essential to note that this scan sent ICMP address mask requests to every valid IP address and waited for a reply. Each ICMP request was sent twice, as we can see in the screenshot below.



Source	Destination	Protocol	Info
10.11.35.214	10.10.68.1	ICMP	Address mask request id=0xa3c4, seq=0/0, ttl=
10.11.35.214	10.10.68.2	ICMP	Address mask request id=0xb793, seq=0/0, ttl=
10.11.35.214	10.10.68.3	ICMP	Address mask request id=0x2d87, seq=0/0, ttl=
10.11.35.214	10.10.68.4	ICMP	Address mask request id=0x091c, seq=0/0, ttl=
10.11.35.214	10.10.68.5	ICMP	Address mask request id=0x692c, seq=0/0, ttl=
10.11.35.214	10.10.68.6	ICMP	Address mask request id=0x4bec, seq=0/0, ttl=
10.11.35.214	10.10.68.7	ICMP	Address mask request id=0x4d61, seq=0/0, ttl=
10.11.35.214	10.10.68.8	ICMP	Address mask request id=0xb84f, seq=0/0, ttl=
10.11.35.214	10.10.68.9	ICMP	Address mask request id=0x7d19, seq=0/0, ttl=
10.11.35.214	10.10.68.10	ICMP	Address mask request id=0x92be, seq=0/0, ttl=
10.11.35.214	10.10.68.1	ICMP	Address mask request id=0xd204, seq=0/0, ttl=
10.11.35.214	10.10.68.2	ICMP	Address mask request id=0x683d, seq=0/0, ttl=
10.11.35.214	10.10.68.3	ICMP	Address mask request id=0x2711, seq=0/0, ttl=
10.11.35.214	10.10.68.4	ICMP	Address mask request id=0xfde3, seq=0/0, ttl=
10.11.35.214	10.10.68.5	ICMP	Address mask request id=0x2eb1, seq=0/0, ttl=
10.11.35.214	10.10.68.6	ICMP	Address mask request id=0x8306, seq=0/0, ttl=

Answer the questions below

What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?

-PP

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?

-PM

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Echo to discover live hosts?

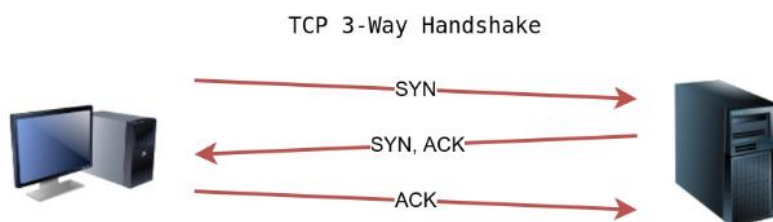
-PE

✓ Correct Answer

Task 7 Nmap Host Discovery Using TCP and UDP

TCP SYN Ping

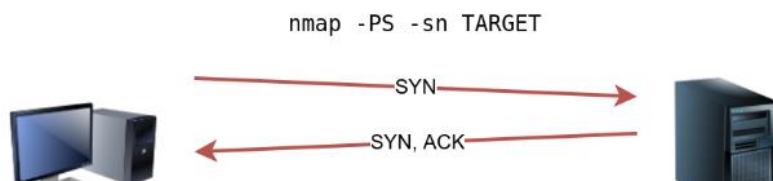
We can send a packet with the SYN (Synchronize) flag set to a TCP port, 80 by default, and wait for a response. An open port should reply with a SYN/ACK (Acknowledge); a closed port would result in an RST (Reset). In this case, we only check whether we will get any response to infer whether the host is up. The figure below is a reminder of how a TCP 3-way handshake usually works.



Case: TCP port is open.

If you want Nmap to use TCP SYN ping, you can do so via the option `-PS` followed by the port number, range, list, or a combination of them. For example, `-PS21` will target port 21, while `-PS21-25` will target ports 21, 22, 23, 24, and 25. Finally `-PS80,443,8080` will target the three ports 80, 443, and 8080.

Privileged users (root and sudoers) can send TCP SYN packets and don't need to complete the TCP 3-way handshake even if the port is open, as shown in the figure below. Unprivileged users have no choice but to complete the 3-way handshake if the port is open.



10.11.35.214	10.10.68.1	UDP	57192	→	40125	Len=40
10.11.35.214	10.10.68.2	UDP	57192	→	40125	Len=40
10.11.35.214	10.10.68.3	UDP	57192	→	40125	Len=40
10.11.35.214	10.10.68.4	UDP	57192	→	40125	Len=40
10.11.35.214	10.10.68.5	UDP	57192	→	40125	Len=40
10.11.35.214	10.10.68.6	UDP	57192	→	40125	Len=40
10.11.35.214	10.10.68.7	UDP	57192	→	40125	Len=40

● nmap-PU-sn-openvpn.pcapng

Packets: 1118 · Displayed: 602 (53.8%) Profile: Default

Masscan

On a side note, Masscan uses a similar approach to discover the available systems. However, to finish its network scan quickly, Masscan is quite aggressive with the rate of packets it generates. The syntax is quite similar: `-p` can be followed by a port number, list, or range. Consider the following examples:

- `masscan MACHINE_IP/24 -p443`
- `masscan MACHINE_IP/24 -p80,443`
- `masscan MACHINE_IP/24 -p22-25`
- `masscan MACHINE_IP/24 --top-ports 100`

Masscan is not installed on the AttackBox; however, it can be installed using `apt install masscan`.

Answer the questions below

Which TCP ping scan does not require a privileged account?

TCP SYN Ping

✓ Correct Answer

Which TCP ping scan requires a privileged account?

TCP ACK Ping

✓ Correct Answer

What option do you need to add to Nmap to run a TCP SYN ping scan on the telnet port?

-PS23

✓ Correct Answer

🔍 Hint

Task 8 Using Reverse-DNS Lookup

Nmap's default behaviour is to use reverse-DNS online hosts. Because the hostnames can reveal a lot, this can be a helpful step. However, if you don't want to send such DNS queries, you use `-n` to skip this step.

By default, Nmap will look up online hosts; however, you can use the option `-R` to query the DNS server even for offline hosts. If you want to use a specific DNS server, you can add the `-dns-servers DNS_SERVER` option.

Answer the questions below

We want Nmap to issue a reverse DNS lookup for all the possible hosts on a subnet, hoping to get some insights from the names. What option should we add?

-R

✓ Correct Answer

Task 9 ✓ Summary

You have learned how ARP, ICMP, TCP, and UDP can detect live hosts by completing this room. Any response from a host is an indication that it is online. Below is a quick summary of the command-line options for Nmap that we have covered.

Scan Type	Example Command
ARP Scan	<code>sudo nmap -PR -sn MACHINE_IP/24</code>
ICMP Echo Scan	<code>sudo nmap -PE -sn MACHINE_IP/24</code>
ICMP Timestamp Scan	<code>sudo nmap -PP -sn MACHINE_IP/24</code>
ICMP Address Mask Scan	<code>sudo nmap -PM -sn MACHINE_IP/24</code>
TCP SYN Ping Scan	<code>sudo nmap -PS22,80,443 -sn MACHINE_IP/30</code>
TCP ACK Ping Scan	<code>sudo nmap -PA22,80,443 -sn MACHINE_IP/30</code>
UDP Ping Scan	<code>sudo nmap -PU53,161,162 -sn MACHINE_IP/30</code>

Remember to add `-sn` if you are only interested in host discovery without port-scanning. Omitting `-sn` will let Nmap default to port-scanning the live hosts.



You did it! 🎉 Nmap Live Host Discovery complete!

Points earned

🎯 160

Completed tasks

📋 9

Room type

🧭 Walkthrough

Difficulty

📶 Medium

Streak

🔥 1

👥 80,407 users are actively learning this week